

# Philip-Chandan Brain Imaging Pipeline

UCSF longitudinal glioma: Phase 0 spike + 7-patient cohort export, expert segmentation, PDE prep

QBIHack brain-cancer-sim | Generated 2026-06-28

Philip-Chandan owns the brain imaging pipeline: download real glioma MRI with expert segmentations (not Otsu heuristics), validate NiftI pairs, export raw 3D numpy arrays, run prepare\_pde\_input.py (resample/crop/normalize to 64^3), and hand solver-ready cubes to Vinesh for PDE growth simulation and calibration. Phase 0 locked UCSF patient 100002; export\_all\_raw.py now batch-exports seven UCSF cohort patients (baseline + follow-up) with checkpointed progress and WT volume QC.

## 1. Phase 0 spike patient

| Field      | Value                          |
|------------|--------------------------------|
| Dataset    | UCSF Longitudinal Glioma       |
| Patient ID | 100002                         |
| Diagnosis  | Oligodendroglioma, WHO grade 2 |
| IDH / MGMT | mut / positive                 |
| Timepoint  | baseline (time1)               |
| Slug       | glioma_ucsf_100002_baseline    |

Unlike breast-cancer-sim (TCIA DICOM + Otsu), brain datasets ship NiftI volumes with radiologist-drawn tumor masks. We use those masks as ground truth for QC and for Vinesh's PDE initial condition per handoff\_contract.json v1.0.0.

## 2. Pipeline modules (Phase 0)

| Module                         | Role                                                                      |
|--------------------------------|---------------------------------------------------------------------------|
| nifti_extractor.py             | Load NiftI as (Z,Y,X) float32; resolve UCSF paths; validate MR+mask pairs |
| export_raw_extract.py          | Write raw .npy + JSON sidecar; copy expert mask to segmentations/         |
| export_all_raw.py              | Checkpointed cohort batch export + optional UCSF workbook volume QC       |
| wt_volume_report.py            | Longitudinal WT mm^3 from expert seg; compare to UCSF Table S1            |
| pde_burden_compare.py          | PDE 64^3 voxel burden vs WT growth % spec                                 |
| qc_slice_plot.py               | Mid-Z PNG + longitudinal before/after overlay panels                      |
| view_volume_napari.py          | Interactive 3D QC viewer with clinical display controls                   |
| cohort/cohort_discovery.py     | Scan local NiftI trees; find longitudinal cases; audit cohort.json        |
| download_mu_glioma_post.py     | TCIA Faspex helper for MU-Glioma-Post backup dataset                      |
| ../vinesh/prepare_pde_input.py | Resample expert mask, normalize MR, crop to 64^3 PDE input                |
| qc_pde_plot.py                 | QC PNGs for resampled expert mask + cropped PDE volume slices             |

### On-disk outputs

| Artifact         | Path                                                                             |
|------------------|----------------------------------------------------------------------------------|
| Raw MR volume    | data/processed/raw-extract-philip-chandan/100002/baseline.npy                    |
| Metadata sidecar | data/processed/raw-extract-philip-chandan/100002/baseline.json                   |
| Expert mask      | data/processed/segmentations/glioma_ucsf_100002_baseline_mask.nii.gz             |
| QC overlay PNG   | data/qc/slice-plots-philip-chandan/glioma_ucsf_100002_baseline_mid-z-overlay.png |
| Longitudinal QC  | data/qc/slice-plots-philip-chandan/{patient_id}_longitudinal_mid-z-overlay.png   |
| PDE prep QC      | data/qc/pde-prep-vinesh/g64/glioma_ucsf_100002_baseline_pde-input-mid-z.png      |
| PDE input        | data/processed/pde-input-vinesh/100002/g64/baseline.npy                          |

### Spike volume metadata

| Property | Value |
|----------|-------|
|----------|-------|

|               |                                                      |
|---------------|------------------------------------------------------|
| Shape (Z,Y,X) | [155, 240, 240]                                      |
| Spacing (mm)  | [1.0, 1.0, 1.0]                                      |
| Source MR     | data/raw/ucsf_alptdg/100002/100002_time1_t1ce.nii.gz |
| Tumor voxels  | 8,137 (expert mask, full volume)                     |

### PDE input metadata

| Property      | Value                                                                |
|---------------|----------------------------------------------------------------------|
| Shape (Z,Y,X) | [64, 64, 64]                                                         |
| Spacing (mm)  | [1.0, 1.0, 1.0]                                                      |
| Segmentation  | expert_mask                                                          |
| Tumor voxels  | 8,137 (cropped 64^3 grid)                                            |
| Source mask   | data/processed/segmentations/glioma_ucsf_100002_baseline_mask.nii.gz |

## 3. Visual QC - expert segmentation overlay

qc\_slice\_plot.py picks the axial slice with the most tumor voxels and draws a lime contour from the UCSF expert mask. This confirms mask alignment before export and before handing data to Vinesh.

glioma\_ucsf\_100002\_baseline z=81

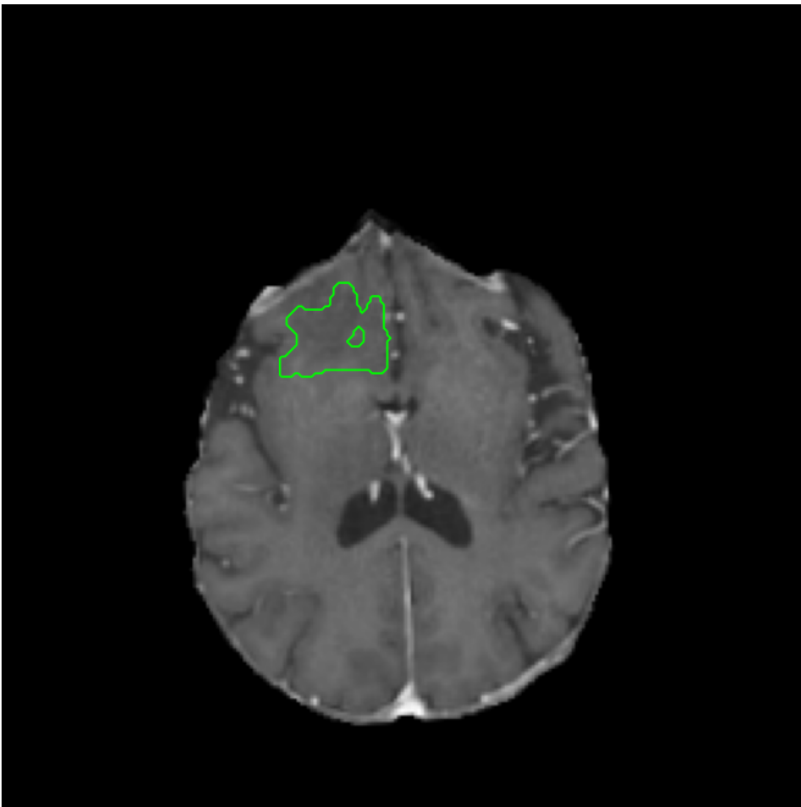


Figure 1. UCSF patient 100002 baseline T1ce with expert segmentation contour (lime). Slice chosen at maximum tumor cross-section.

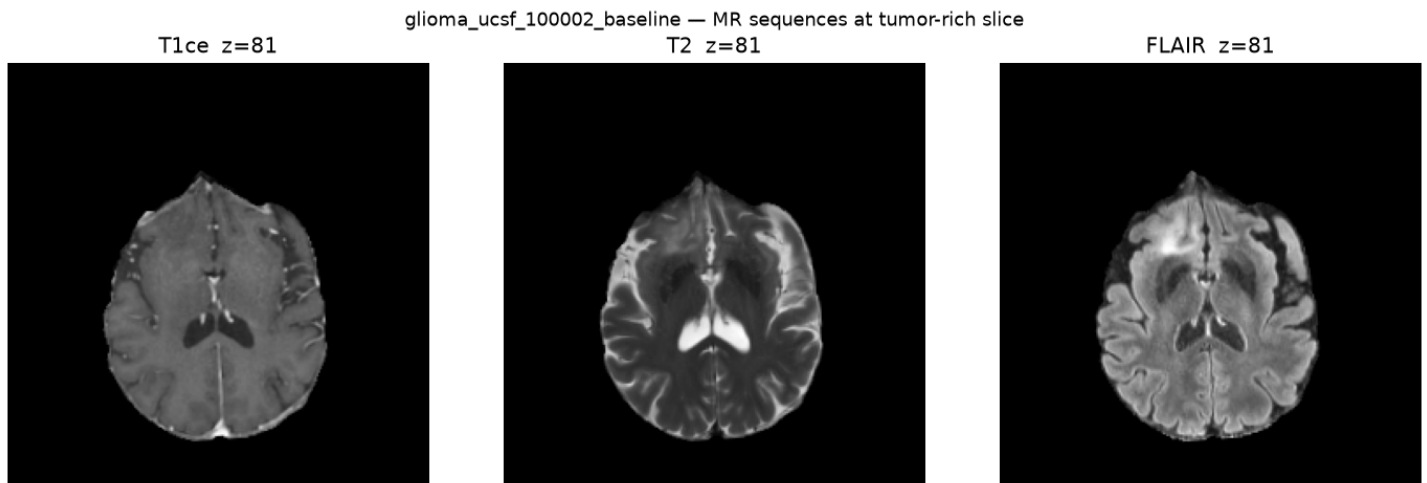


Figure 2. Same z slice across MR sequences available for this timepoint. Low-grade gliomas may show subtle T1ce enhancement; FLAIR often highlights edema.

## 4. MRI sequences - what T1, T2, FLAIR, and T1ce mean

Hospital brain MRI protocols acquire several contrasts of the same anatomy. Radiologists read them together; our napari viewer loads all UCSF series for the spike timepoint so you can toggle layers like on a PACS workstation.

| Sequence | Full name                           | What you see                                             | Glioma relevance                                            |
|----------|-------------------------------------|----------------------------------------------------------|-------------------------------------------------------------|
| T1       | T1-weighted                         | Anatomy; CSF dark, fat bright                            | Baseline anatomy; pre-contrast reference                    |
| T1ce     | T1 post-contrast (T1+C)             | Areas with leaky blood vessels brighten after gadolinium | Enhancing tumor rim; spike export uses T1ce as primary MR   |
| T2       | T2-weighted                         | Fluid bright; good soft-tissue contrast                  | Tumor and edema often appear brighter than normal brain     |
| FLAIR    | Fluid-attenuated inversion recovery | T2-like but CSF suppressed (dark ventricles)             | Peritumoral edema and infiltrative signal stand out clearly |

For our grade 2 IDH-mut oligodendroglioma spike, T1ce enhancement is subtle - mean intensity inside the expert mask is almost identical to surrounding brain. FLAIR is often more informative for edema. The expert mask delineates the lesion using all available clinical context, not brightness on a single sequence.

## 5. Clinical napari viewer - view\_volume\_napari.py

We extended the napari QC viewer beyond a simple grayscale overlay. Controls live in the right dock panels. CLAHE and window/level adjustments are labeled QC-only - they help engineers inspect data, not replace diagnostic PACS reading.

### Window / Level (WW/WL)

Radiology displays map MR intensities to gray using a window (width) and level (center). Narrow windows increase contrast but clip bright/dark regions. Our defaults compute level and width from brain voxels only (2nd-98th percentile), ignoring black padding outside the head. Sliders update all MR layers; hold Ctrl and drag on the canvas for click-drag WW/WL like a clinical viewer.

### Multi-sequence layers

For UCSF patient folders, the viewer auto-loads T1, T1ce, T2, and FLAIR when Nifti files exist. T1ce is visible by default with the expert overlay. Toggle FLAIR or T2 in napari's layer list to compare edema vs enhancement on the same slice.

## Plane selector and orthogonal MPR grid

The Plane dropdown switches axial, coronal, or sagittal viewing on the main canvas while keeping the segmentation aligned. Orthogonal MPR grid opens a 1x3 linked layout (axial | coronal | sagittal) for navigating the tumor in three planes; the overlay is hidden in grid mode because napari assigns one layer per grid cell.

## CLAHE (Contrast Limited Adaptive Histogram Equalization)

CLAHE enhances local contrast slice-by-slice inside the brain mask. It can make subtle texture differences easier to see during QC. It is not a physical imaging sequence and must not be used for diagnosis or as PDE input - it is a display filter only, clearly marked in the viewer.

## Overlay toggle

Hide overlay / Show overlay toggles the expert segmentation labels so you can inspect underlying MR intensity. For subtle T1ce lesions, pairing this with FLAIR and WW/WL adjustment is the intended workflow.

## Launch command

```
cd brain-cancer-sim
.venv/bin/python simulation-vinesh-philip-chandan/philip-chandan/view_volume_napari.py --slug glioma_ucsf_100002_baseline
```

## 6. Expert-mask PDE prep - prepare\_pde\_input.py (Philip-Chandan)

Philip-Chandan runs `prepare_pde_input.py` after each raw export, writing PDE-ready volumes to `data/processed/pde-input-vinesh/<patient_id>/g64/` per `handoff_contract.json v1.0.0`. Unlike `breast-cancer-sim` (Otsu on DCE-MRI), `brain v1` uses the dataset expert mask - no Otsu fallback. Vinesh consumes these cubes for `run_growth()` and `calibrate.py`.

- Resample MR (linear) and expert mask (nearest) to isotropic 1 mm spacing
- Min-max normalize MR intensities to [0, 1]
- Keep continuous normalized density inside the expert mask; zero background
- Crop/pad to max  $64^3$  centered on mask center of mass

Continuous tumor values (not a binary mask) are required so the PDE logistic growth term  $\rho \cdot u \cdot (1-u)$  can evolve. `qc_pde_plot.py` documents the same pipeline via `prepare_pde_stages()` for the PDF figures below.

100002 g64 expert mask z=81 (mask vox=8137)

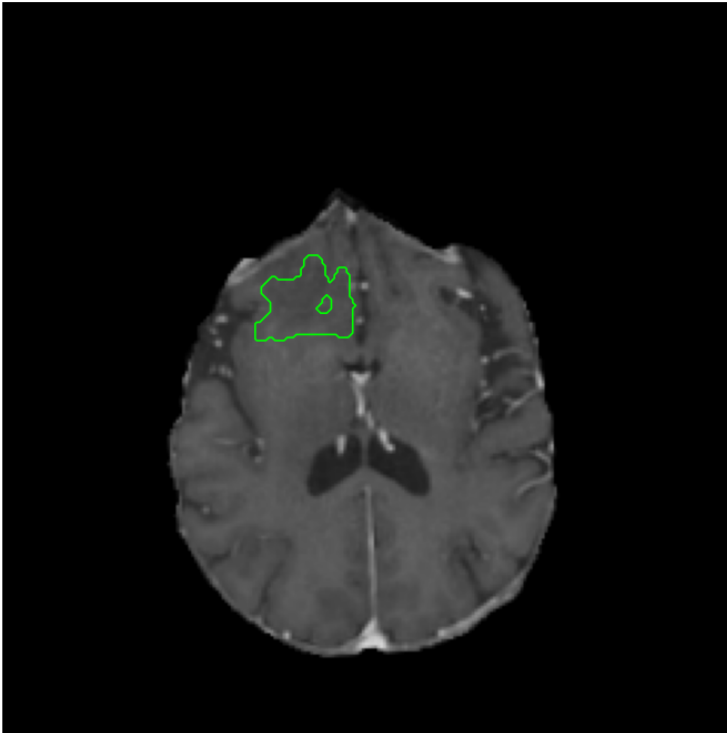


Figure 3. UCSF patient 100002 after resample + normalize. Lime contour is the expert tumor mask on the 1 mm grid before crop; background will be zeroed.

Patient 100002 g64 cropped PDE input (inferno, cyan contour > 0)  
 Baseline (t1) z=30 shape=[64, 64, 64]      Follow-up (t2) z=29 shape=[64, 64, 64]

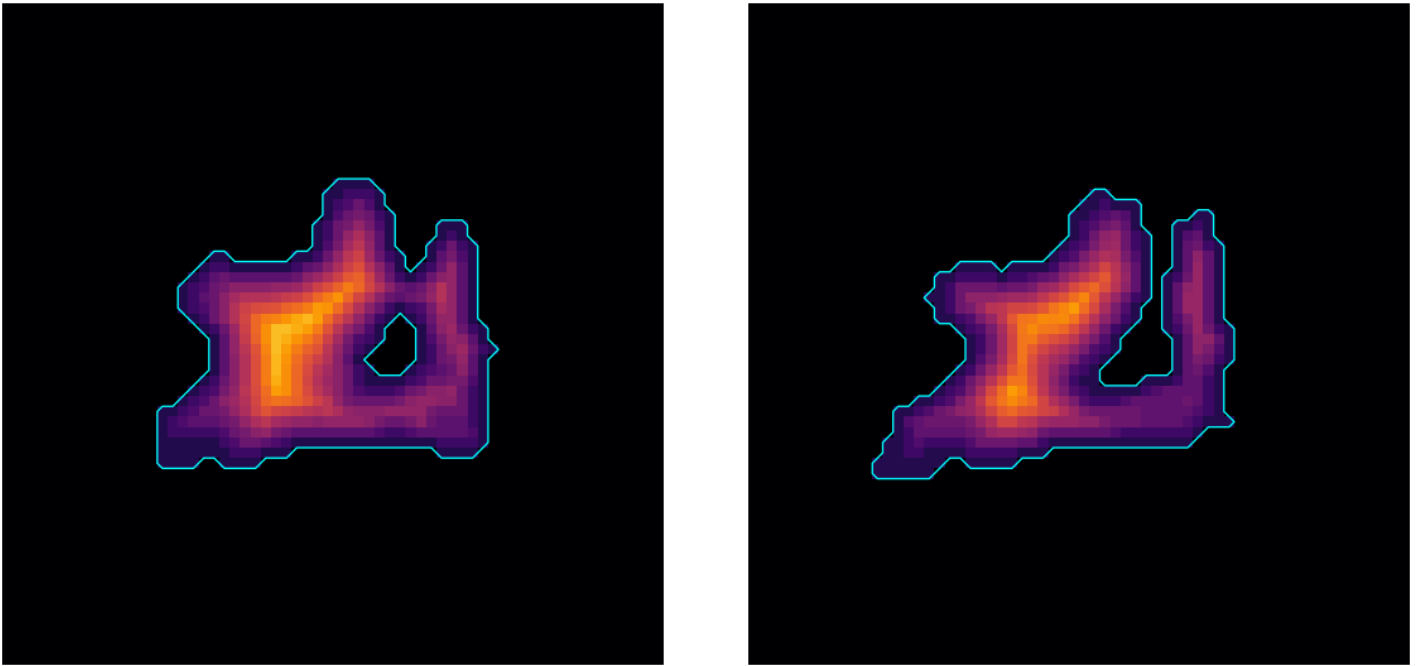


Figure 4. Cropped PDE input ( $64^3$ , 1 mm) for spike patient 100002. Left: baseline (t1); right: follow-up (t2). Inferno colormap shows continuous initial tumor burden; cyan contour marks voxels  $> 0$ .

### Solver smoke test on spike PDE input

run\_growth() on glioma\_ucsf\_100002\_baseline: 51 frames, shape [64, 64, 64], tumor volume ( $u > 0$ ) at  $t=0$  = 262144  $\text{mm}^3$ , at  $t=\text{end}$  ( $u \geq 0.5$ ) = 5738  $\text{mm}^3$ . Growth term is active because initial burden is continuous inside the expert ROI.

### Regenerate PDE artifacts

```
cd brain-cancer-sim
.venv/bin/python simulation-vinesh-philip-chandan/vinesh/prepare_pde_input.py
.venv/bin/python simulation-vinesh-philip-chandan/philip-chandan/qc_pde_plot.py
```

## 7. PDE burden vs WT growth spec (QC)

After prepare\_pde\_input.py, tumor burden in the  $64^3$  PDE cube is the count of voxels with value  $> 0$  (handoff\_contract tumor\_burden\_rule). At 1 mm spacing each voxel is  $\sim 1 \text{ mm}^3$  when the lesion fits inside the crop. WT volumes use BraTS labels 1+2+3 only. Five cohort patients (100130, 100134, 100192, 100220, 100260) have label 4 resection cavity on at least one visit and are omitted below - PDE prep seeds mask  $> 0$ , which inflates capture vs WT. Tables show the two patients without resection cavity (100002 spike + 100118 demo contrast).

### Per-timepoint burden capture (WT $\text{mm}^3$ vs PDE voxels)

| Patient | TP | WT $\text{mm}^3$ | PDE voxels | Capture |
|---------|----|------------------|------------|---------|
| 100002  | t1 | 8,137            | 8,137      | 100.0%  |
| 100002  | t2 | 8,365            | 8,365      | 100.0%  |
| 100014  | t1 | 61,814           | 58,694     | 95.0%   |
| 100014  | t2 | 36,894           | 34,384     | 93.2%   |
| 100020  | t1 | 62,770           | 49,518     | 78.9%   |
| 100020  | t2 | 105,601          | 91,568     | 86.7%   |
| 100038  | t1 | 43,342           | 43,342     | 100.0%  |

|        |    |         |         |        |
|--------|----|---------|---------|--------|
| 100038 | t2 | 78,774  | 76,912  | 97.6%  |
| 100039 | t1 | 84,326  | 80,944  | 96.0%  |
| 100039 | t2 | 64,638  | 63,418  | 98.1%  |
| 100118 | t1 | 10,905  | 10,881  | 99.8%  |
| 100118 | t2 | 77,327  | 72,755  | 94.1%  |
| 100129 | t1 | 1,641   | 1,640   | 99.9%  |
| 100129 | t2 | 1,606   | 1,606   | 100.0% |
| 100137 | t1 | 50,346  | 38,113  | 75.7%  |
| 100137 | t2 | 49,255  | 36,915  | 74.9%  |
| 100149 | t1 | 26,623  | 20,986  | 78.8%  |
| 100149 | t2 | 31,798  | 22,244  | 70.0%  |
| 100184 | t1 | 37,508  | 25,145  | 67.0%  |
| 100184 | t2 | 83,910  | 69,009  | 82.2%  |
| 100210 | t1 | 21,400  | 21,400  | 100.0% |
| 100210 | t2 | 38,832  | 38,329  | 98.7%  |
| 100213 | t1 | 68,879  | 60,137  | 87.3%  |
| 100213 | t2 | 133,812 | 104,443 | 78.1%  |
| 100215 | t1 | 3,284   | 3,284   | 100.0% |
| 100215 | t2 | 2,933   | 2,933   | 100.0% |
| 100219 | t1 | 68,397  | 46,142  | 67.5%  |
| 100219 | t2 | 83,146  | 49,048  | 59.0%  |

### Longitudinal growth % - WT spec vs PDE voxels

| Patient | Cap t1 | Cap t2 | WT d%   | PDE d%  | d diff | QC flags                                                      |
|---------|--------|--------|---------|---------|--------|---------------------------------------------------------------|
| 100002  | 100.0% | 100.0% | +2.8%   | +2.8%   | +0.0   | ok                                                            |
| 100014  | 95.0%  | 93.2%  | -40.3%  | -41.4%  | -1.1   | crop_loss_baseline,<br>crop_loss_followup                     |
| 100020  | 78.9%  | 86.7%  | +68.2%  | +84.9%  | +16.7  | crop_loss_baseline,<br>crop_loss_followup,<br>growth_mismatch |
| 100038  | 100.0% | 97.6%  | +81.7%  | +77.5%  | -4.3   | ok                                                            |
| 100039  | 96.0%  | 98.1%  | -23.3%  | -21.7%  | +1.7   | ok                                                            |
| 100118  | 99.8%  | 94.1%  | +609.1% | +568.6% | -40.5  | crop_loss_followup,<br>growth_mismatch                        |
| 100129  | 99.9%  | 100.0% | -2.1%   | -2.1%   | +0.1   | ok                                                            |
| 100137  | 75.7%  | 74.9%  | -2.2%   | -3.1%   | -1.0   | crop_loss_baseline,<br>crop_loss_followup                     |
| 100149  | 78.8%  | 70.0%  | +19.4%  | +6.0%   | -13.4  | crop_loss_baseline,<br>crop_loss_followup,<br>growth_mismatch |
| 100184  | 67.0%  | 82.2%  | +123.7% | +174.4% | +50.7  | crop_loss_baseline,<br>crop_loss_followup,<br>growth_mismatch |
| 100210  | 100.0% | 98.7%  | +81.5%  | +79.1%  | -2.4   | ok                                                            |
| 100213  | 87.3%  | 78.1%  | +94.3%  | +73.7%  | -20.6  | crop_loss_baseline,<br>crop_loss_followup,<br>growth_mismatch |
| 100215  | 100.0% | 100.0% | -10.7%  | -10.7%  | +0.0   | ok                                                            |

|        |       |       |        |       |       |                                                               |
|--------|-------|-------|--------|-------|-------|---------------------------------------------------------------|
| 100219 | 67.5% | 59.0% | +21.6% | +6.3% | -15.3 | crop_loss_baseline,<br>crop_loss_followup,<br>growth_mismatch |
|--------|-------|-------|--------|-------|-------|---------------------------------------------------------------|

Spike patient 100002 matches exactly (100% capture, +2.8% growth). 0/14 non-resection patients pass all QC flags (capture 95-105% both timepoints and growth within +/-5% of WT spec). Patient 100118 shows follow-up crop loss on the 64^3 grid - Vinesh calibrate.py should use full-res WT targets, not cropped voxel counts alone. Full cohort JSON (all 7 patients) sidecar: data/processed/pde-input-vinesh/pde\_burden\_compare.json.

## 8. UCSF cohort scale-up - WT volume QC (no resection cavity)

export\_all\_raw.py exported baseline + follow-up raw extracts for seven UCSF-LPTDG patients in cohort.json. Five have label 4 resection cavity (excluded from tables below). Longitudinal T1ce overlay PNGs for all seven are on disk under data/qc/slice-plots-philip-chandan/. WT volumes (labels 1+2+3) from wt\_volume\_report.py; use --compare-ucsf-workbook on export\_all\_raw to diff against UCSF Table S1.

| Patient | Grade | IDH | WT t1 (mm3) | WT t2 (mm3) | d mm3   | d %     | Days |
|---------|-------|-----|-------------|-------------|---------|---------|------|
| 100002  | 2     | mut | 8,137       | 8,365       | +228    | +2.8%   | 183  |
| 100014  | 2.0   | WT  | 61,814      | 36,894      | -24,920 | -40.3%  | 53   |
| 100020  | 4.0   | WT  | 62,770      | 105,601     | +42,831 | +68.2%  | 72   |
| 100038  | 4.0   | WT  | 43,342      | 78,774      | +35,432 | +81.7%  | 63   |
| 100039  | 4.0   | WT  | 84,326      | 64,638      | -19,688 | -23.3%  | 27   |
| 100118  | 4.0   | WT  | 10,905      | 77,327      | +66,422 | +609.1% | 56   |
| 100129  | 2.0   | mut | 1,641       | 1,606       | -35     | -2.1%   | 109  |
| 100137  | 2.0   | mut | 50,346      | 49,255      | -1,091  | -2.2%   | 71   |
| 100149  | 4.0   | WT  | 26,623      | 31,798      | +5,175  | +19.4%  | 56   |
| 100184  | 3.0   | mut | 37,508      | 83,910      | +46,402 | +123.7% | 105  |
| 100210  | 4.0   | WT  | 21,400      | 38,832      | +17,432 | +81.5%  | 60   |
| 100213  | 3.0   | WT  | 68,879      | 133,812     | +64,933 | +94.3%  | 57   |
| 100215  | 3.0   | WT  | 3,284       | 2,933       | -351    | -10.7%  | 63   |
| 100219  | 2.0   | WT  | 68,397      | 83,146      | +14,749 | +21.6%  | 37   |

## 9. Handoff contract summary

| Philip-Chandan (raw extract)       | Vinesh (PDE input)                              |
|------------------------------------|-------------------------------------------------|
| Raw MR intensities, not normalized | Philip-Chandan: prepare_pde_input -> 64^3, 1 mm |
| spacing_mm in JSON sidecar         | Normalize to [0, 1] inside tumor region         |
| Expert mask path in metadata       | Expert mask resampled - no Otsu fallback        |
| Axis order (Z, Y, X)               | Vinesh: run_growth() + calibrate.py             |

## 10. Summary and next steps

Phase 0 delivered NiftI ingest, UCSF spike export, expert-mask QC, napari viewer, Philip-Chandan PDE prep into 64^3 continuous-density input, and solve\_growth() smoke on patient 100002. Cohort scale-up exported seven patients (14 timepoints); PDE burden and WT tables in this PDF cover the two without resection cavity (100002, 100118). Next: manifest.json v1.0.0, Vinesh calibrate.py for t1->t2, and demo toggle (100002 stable IDH-mut vs 100118 aggressive IDH-WT GBM).

### Deliverables completed

| Stage | Deliverable | Status |
|-------|-------------|--------|
|-------|-------------|--------|



|                 |                                                   |                           |
|-----------------|---------------------------------------------------|---------------------------|
| Cohort          | cohort.json rev1 - 7 UCSF patients                | Done                      |
| Export          | export_all_raw.py checkpointed batch              | 7 patients × 2 timepoints |
| Volume QC       | wt_volume_report.json + workbook compare          | Done (7 exported)         |
| Longitudinal QC | 7 before/after overlay PNGs                       | On disk                   |
| PDE prep        | prepare_pde_input g64 - 7 patients x 2 timepoints | Done                      |
| PDE burden QC   | pde_burden_compare.json + section 7               | 2 non-resection in PDF    |
| Solver          | run_growth() smoke on 100002 baseline             | Passed                    |
| Manifest        | manifest.json v1.0.0 (2 patients)                 | Done                      |
| Pending         | calibrate.py, Jasim/Vinesh/Philip                 | Next                      |